

TEXTILE SALES ANALYTICS PROJECT

End-to-End Documentation: Excel Cleaning • SQL Analysis • Power BI
Dashboard



*Prepared as a project documentation summary
Dataset Period: Jan 2026 – Dec 2026*

1. Project Overview

This project analyses textile trade sales data through a complete analytics pipeline — raw data cleaning in Excel, exploratory and aggregate analysis using SQL Server, and an interactive Power BI dashboard for business reporting.

Objectives

- Clean and standardize raw textile sales records (600 orders, 25 fields).
- Fix inconsistent categorical entries, negative quantities, and mismatched sales values.
- Run SQL-based analysis: totals, rankings, growth trends, and status breakdowns.
- Build a Power BI dashboard for interactive, visual business reporting.
- Derive actionable insights on region, product, customer, and payment performance.

Tools & Technology Stack

Stage	Tool	Purpose
Data Cleaning	Microsoft Excel	Fix raw fields, handle nulls/errors, standardize text
Data Analysis	SQL Server (T-SQL)	Aggregation, ranking, growth & trend queries
Visualization	Power BI (.pbit)	Interactive dashboard & KPI reporting

2. Data Source & Structure

The dataset "Sales_cleaned.csv" contains 600 textile trade orders across 12 Indian states, 10 products, and 12 customers, spanning Jan–Dec 2026. It retains both raw and cleaned versions of key fields for audit traceability.

Field Group	Columns
Order Info	Order_ID, Order_Date, Order_Month, Order_Quarter, Order_Year
Customer & Location	Customer_ID, Customer_Name, State, Region_Raw / Region
Product	Product_ID, Product_Name, Fabric_Type
Quantity & Price	Quantity_Meters_Raw / Quantity_Meters, Unit_Price, Discount_Percent
Sales Value	Gross_Amount, Sales_Amount_Raw / Sales_Amount_Expected, Sales_Match
Status	Order_Status_Raw / Order_Status, Payment_Method_Raw / Payment_Method

Dataset Snapshot

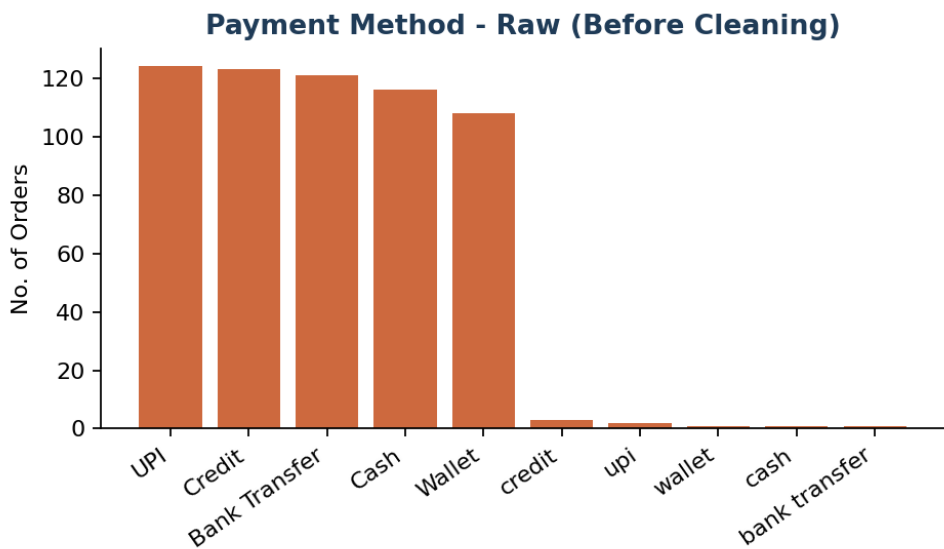
Metric	Value
Total Records	600
Total Fields	25
Date Range	Jan 2026 – Dec 2026
States Covered	12
Product Lines	10
Unique Customers	12

3. Data Cleaning Process (Excel)

Raw data contained typical real-world quality issues. Each was identified and corrected, keeping the original "_Raw" column for traceability.

Issue Found	Column	Fix Applied
Mixed case text (delivered, processing)	Order_Status	Standardized to Title Case (Delivered, Processing)
Mixed case & abbreviations (upi, cash, credit)	Payment_Method	Standardized to consistent labels (UPI, Cash, Credit, etc.)
Missing values (5 blanks) & error tags (#REF!)	Region	Flagged for correction; mapped from State where possible
Negative values (4 records)	Quantity_Meters	Converted to absolute / corrected positive values
Raw vs expected mismatch check	Sales_Amount	Validated Gross Amount – Discount = Expected Sales (Sales_Match flag)

Payment Method — Before Cleaning (Raw Text Variants)



10 inconsistent text variants were consolidated into 5 standardized payment categories.

4. SQL Analysis Performed

Analysis was performed on SQL Server against the [Textile_Sales].[dbo].[Textile] table, covering summary metrics, grouped aggregates, ranking, and trend functions.

Category	Queries Performed
Summary Metrics	Total order count, total sales, average unit price, max/min order value
Grouped Aggregates	Sales by Region, State, Fabric Type, Customer, Product
Top-N Analysis	Top 10 customers by sales, Top 10 products by sales
Status Breakdown	Order count by Payment Method and Order Status
Time-based Trends	Monthly sales, monthly order count, month-over-month growth %
Window Functions	RANK() for customer sales, DENSE_RANK() for product quantity, LAG() for growth %
Conditional Logic	CASE-based Delivered vs Non-Delivered split; Cancelled/Returned % calculation
Data Validation	HAVING clause to find customers with >20 orders
Reusability	CREATE VIEW Cleaned_Textile_Sales_Data for repeatable reporting access

Sample Query — Month-over-Month Growth

```
SELECT Order_Year, Order_Month, ((SUM(Sales_Amount_Expected) - LAG(SUM(Sales_Amount_Expected)) OVER(...)) / LAG(...)) * 100 AS Growth_Percentage FROM Textile GROUP BY Order_Year, Order_Month ORDER BY Order_Year, Month;
```

5. Key Business Metrics (KPI Summary)

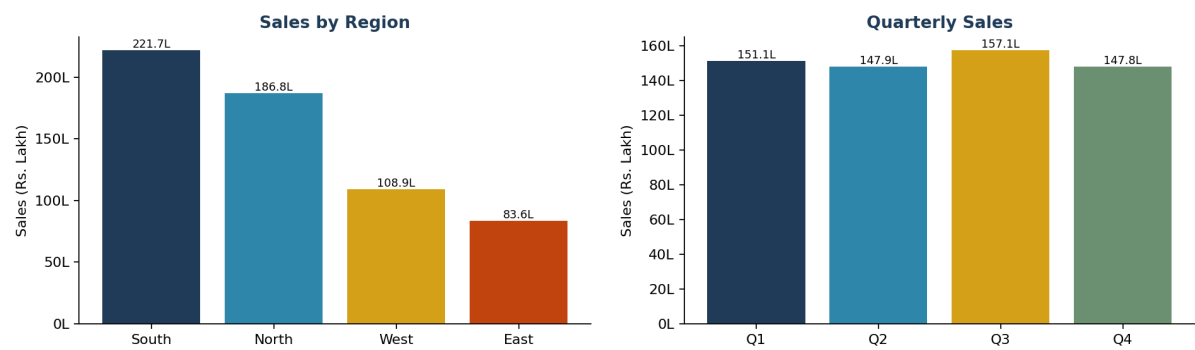


Highlights

- South region leads with Rs. 2.22 Cr in sales (~37% of total revenue).
- Silk Fusion is the top-performing product, generating Rs. 1.31 Cr alone.
- Vardhan Textiles is the highest-value customer at Rs. 62.4 L in total purchases.
- Delivered orders account for 55% of all orders (328 of 600).
- Payment methods are evenly distributed — UPI and Credit lead at 126 orders each.

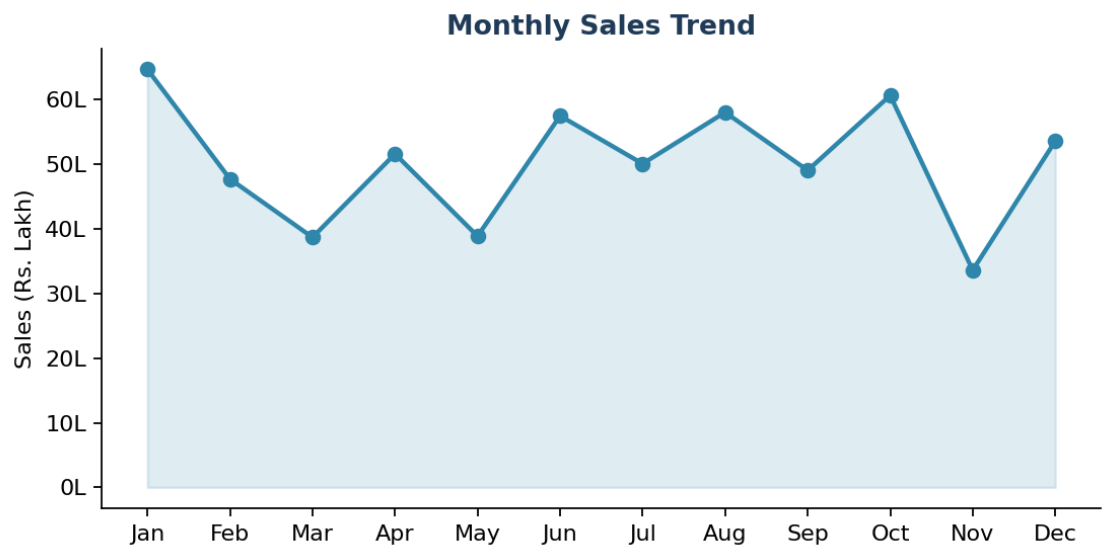
6. Visual Analysis

6.1 Regional & Quarterly Sales



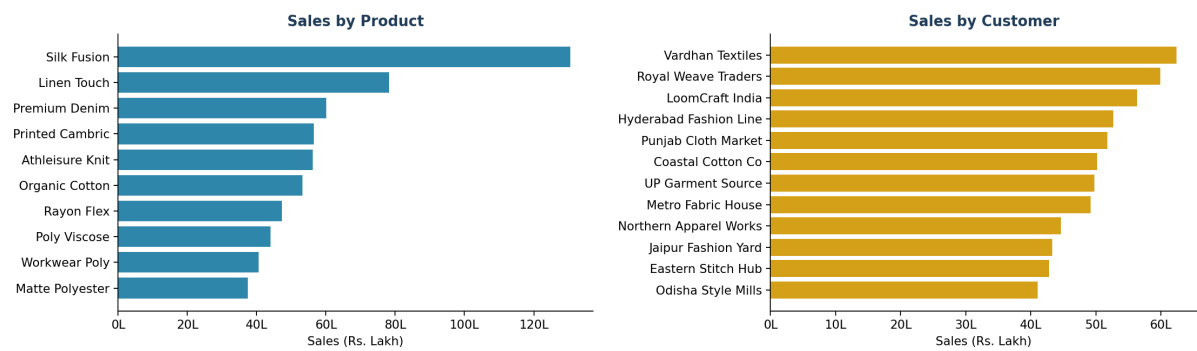
South region dominates overall sales; quarterly revenue is fairly stable (Rs. 1.48–1.57 Cr per quarter), indicating consistent demand year-round.

6.2 Monthly Sales Trend



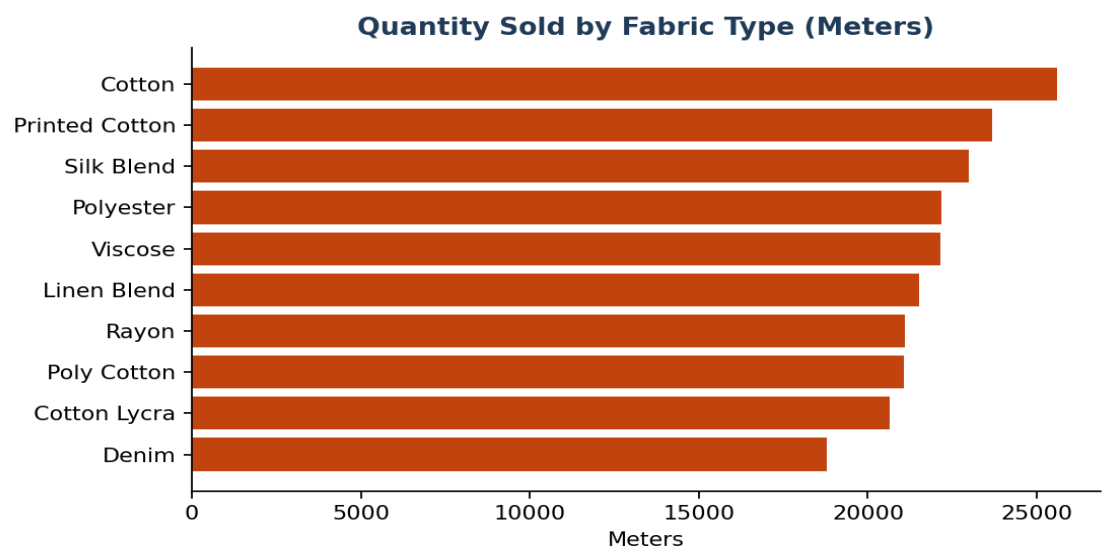
January and October are peak months; November dips lowest, likely a seasonal/order-cycle effect.

6.3 Product & Customer Performance



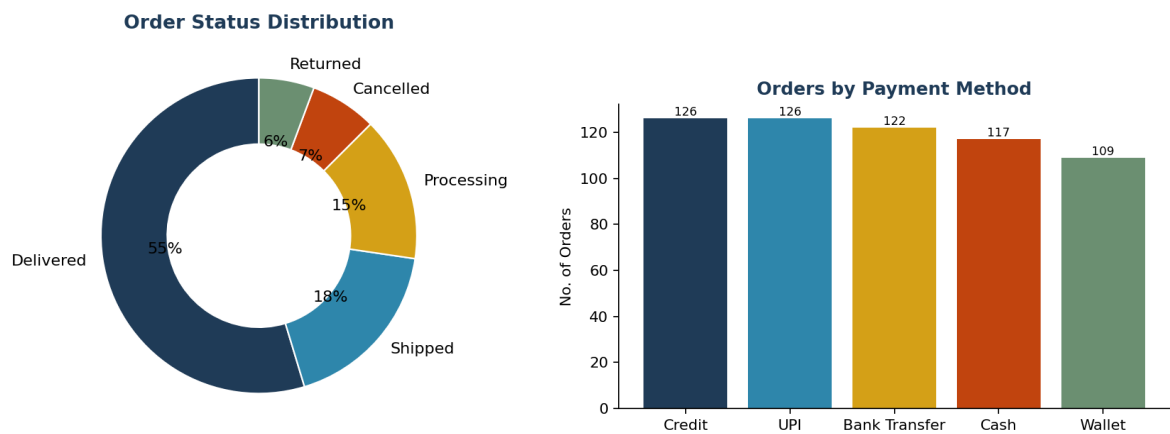
Silk Fusion outsells the next product by nearly 2x. Top 3 customers together contribute over Rs. 1.78 Cr, ~30% of total revenue.

6.4 Fabric Demand (Quantity in Meters)



Cotton and Printed Cotton are the most in-demand fabrics by volume, though not the highest by revenue — indicating a lower average price point.

6.5 Order Status & Payment Method



Cancelled + Returned orders make up 12.5% of total — a key area for operational review. Payment method usage is well balanced across all five channels.

7. Power BI Dashboard

The Power BI template (Textile_PowerBI.pbix) connects to the cleaned dataset and consolidates the above metrics into a single interactive report for business users.

Dashboard Components

- KPI Cards: Total Sales, Total Orders, Average Unit Price, Cancelled/Returned %.
- Regional map / bar chart of sales by Region and State.
- Trend line chart for monthly and quarterly sales performance.
- Top-N visuals for best-selling Products and top Customers.
- Donut/pie chart for Order Status and Payment Method distribution.
- Slicers for Region, Fabric Type, Order Status, and Month for interactive filtering.

Data Model

The report uses a single fact table (Cleaned Textile Sales) built on the SQL view **Cleaned_Textile_Sales_Data**, ensuring the dashboard always reflects validated, de-duplicated, and standardized records rather than raw source data.

8. Key Insights & Recommendations

Insights

- Revenue is concentrated: South + North regions contribute ~68% of total sales — East and West are under-penetrated.
- Silk Fusion and Linen Touch together drive ~34% of revenue — high-value, low-volume products.
- Cotton-based fabrics dominate by volume but not by value — opportunity to upsell premium fabrics.
- 12.5% cancellation/return rate is above a healthy 5–8% benchmark and merits root-cause review.
- Sales are evenly spread across payment methods, showing no channel dependency risk.

Recommendations

- Investigate cancellation/return drivers (e.g. Order_Status_Raw noted 7 raw sub-statuses before cleanup).
- Run targeted marketing in East & West regions to balance regional demand.
- Bundle high-volume cotton fabrics with premium products (Silk, Linen) to lift average order value.
- Automate the Excel cleaning rules (case standardization, negative-quantity fix) into the SQL ETL/view layer.
- Refresh the Power BI report on a monthly cadence aligned with the Order_Month cycle.

9. Conclusion

This project demonstrates a complete analytics workflow — from raw, inconsistent transactional data to a validated SQL data layer and an interactive Power BI dashboard. The cleaning stage resolved case inconsistencies, missing regions, negative quantities, and value mismatches, producing a single reliable source of truth (Cleaned_Textile_Sales_Data view). SQL analysis surfaced ranking, growth, and status-level insights, which are now visualized for ongoing business monitoring via Power BI.

Project Summary

Stage	Deliverable
Cleaning	Sales_cleaned.csv — 600 records, 25 fields, fully standardized
Analysis	SQLQueries.sql — 25 queries incl. aggregates, ranks, growth, view
Reporting	Textile_PowerBI.pbit — interactive dashboard template

— End of Documentation —